This is not the ex-gaussian model you are looking for: Commentatry on the Misuse of Bayesian ex-Gaussian models in {brms}

Jason Geller[*]

Boston College, Department of Psychology and Neuroscience

2026-07-06

The ex-Gaussian distribution is a popular tool for analyzing reaction-time (RT) data because it distinguishes the central portion of the distribution, summarized by the Gaussian parameters μ and σ, from its positively skewed tail, summarized by the exponential parameter τ. Bayesian ex-Gaussian models can now be fit easily with the {brms} package in R, but the default parameterization contains a subtle pitfall: the parameter labeled `mu` represents the mean of the full ex-Gaussian distribution, $E(RT) = \mu + \tau$, rather than the location of the Gaussian component. Consequently, effects estimated on the default `mu` describe changes in the overall mean and can obscure genuine shifts in the Gaussian component. Using simulated data in which the Gaussian location and the exponential tail move in opposite directions—holding the overall mean constant—I show that the default model returns a null effect on `mu`, whereas a classical parameterization recovers the true 50-ms shift. I provide annotated {brms} code for both parameterizations and argue that convenient software defaults should not replace careful attention to what a model's parameters actually estimate.

Keywords: ex-Gaussian, reaction time, Bayesian estimation, brms, parameterization
Words: 2798

Over the past two decades, psychology has seen a substantial increase in the application of Bayesian data analysis methods as a supplement to widely used frequentist methods (Pfadt et al., 2025). This shift has been driven by the increasing availability of computational tools that have made fitting Bayesian models both practical and efficient. Programming languages like R (R Core Team, 2024), Python (Van Rossum & Drake, 2009), and Julia (Bezanson et al., 2017) provide libraries and packages that allow researchers to fit Bayesian models with little technical overhead. In addition, growing dissatisfaction with null-hypothesis significance testing (NHST) and its associated pitfalls (e.g., p-value misinterpretation and dichotomous thinking), together with the replication crisis and credibility revolution (Vazire, 2018), has pushed many researchers toward statistical alternatives.

One domain that has seen an increase in the adoption of Bayesian methods is the analysis of response time (RT) data, where RTs are routinely used as measures of processing speed, decision difficulty, and cognitive efficiency. Traditionally, RT data have often been analyzed using summary statistics such as the mean or median. However, RT distributions are typically positively skewed, bounded at zero, and characterized by substantial variability both within and between individuals (Wilcox & Keselman, 2003). Moreover, experimental manipulations may affect not only the central tendency of the distribution but also its variability or upper tail. Consequently, analyses based solely on the mean or median can obscure theoretically meaningful distributional effects, motivating calls to move beyond mean-based analyses of RT data (Balota & Yap, 2011).

One way to move beyond the mean is to model the full shape of the RT distribution. Mathematical models provide a flexible way to do so, allowing researchers to draw inferences about the psychological processes—such as memory, attention, and language—that give rise to observed response times. One model that has become especially popular for this purpose is the ex-Gaussian model (Ratcliff & Murdock, 1976). The ex-Gaussian distribution is formed by the convolution of a Gaussian distribution and an exponential distribution and is characterized by three parameters: μ,

[*]Send correspondence to: Jason Geller, drjasongeller@gmail.com.

σ, and τ. The parameter μ represents the mean of the Gaussian component, whereas σ represents its standard deviation. The parameter τ represents both the mean and standard deviation of the exponential component and primarily governs the length of the distribution's positively skewed upper tail. The overall mean of the ex-Gaussian distribution is $\mu + \tau$, and its variance is $\sigma^2 + \tau^2$. Although these parameters have been used to infer cognitive processes such as automaticity and cognitive control/inhibition, these interpretations are not inherent to the parameters and should be made cautiously (Matzke & Wagenmakers, 2009).

The ex-Gaussian model can be fit within a Bayesian framework relatively easily using the {brms} package in R which provides a high-level interface to Stan (Carpenter et al., 2017). The package supports a wide range of response distributions, including the ex-Gaussian family, and allows researchers to incorporate complex model structures at both the participant and item levels as well as across different parameters. This accessibility has contributed to the growing use of Bayesian ex-Gaussian models in reaction-time research. However, an important complication arises from the default parameterization used by {brms}. Although the model is straightforward to estimate, its default specification does not necessarily provide direct access to the parameter-specific comparisons researchers often wish to make.

The central problem is that the parameter labeled μ in the default {brms} ex-Gaussian model does not correspond to the mean of the Gaussian component in the conventional parameterization. Instead, the quantity modeled as μ in {brms} represents the mean of the full ex-Gaussian distribution. Under the conventional parameterization, this quantity is equal to $\mu_{\text{Gaussian}} + \tau$. Consequently, regression effects estimated for the default μ parameter reflect changes in a composite quantity that combines the Gaussian location and exponential tail components. Such effects therefore cannot be interpreted directly as changes in the Gaussian component or in typical response speed. To recover the conventional Gaussian mean, it is necessary to derive it from the model parameters as $\mu_{\text{Gaussian}} = \mu_{\text{brms}} - \tau$ or to specify an alternative parameterization that models the conventional ex-Gaussian parameters directly.

In this commentary, I demonstrate what the default parameterization of the ex-Gaussian model in **brms** actually provides and then show how to specify the model so that its parameters correspond to the conventional ex-Gaussian parameters used in reaction-time research. The goal is to improve the accuracy and interpretability of inferences drawn from these models. This issue is particularly important because the ex-Gaussian distribution is widely used to distinguish changes in the central portion of the RT distribution from changes in its posietively skewed tail.

Example

To demonstrate the issue with fitting the default ex-Gaussian model in {brms}, I simulate data using a comparison adapted from one of my own studies (Geller et al., 2025) in R (R Core Team (2024); version 4.6.0). This allows the methodological issue to be illustrated without singling out any particular researchers or journals, although published examples of this interpretive problem do exist.

In Geller et al. (2025), we examined how perceptual disfluency—manipulated by varying the level of blur applied to words—influenced reaction times in a lexical decision task (i.e., is this letter string a word?), and whether these effects were related to recognition memory. In that study, low blur affected early stages of encoding, reflected in the Gaussian location parameter μ, but not the later, tail component captured by τ. For the present illustration, however, I construct a sharper scenario in which μ and τ differ in opposite directions so that the two conditions share the same expected reaction time. Using the `rexgaussian2()` function from the {brms.exgaussian} package, I simulate ex-Gaussian data for clearly presented and low-blur words in which μ is set to 500 ms and 550 ms, τ to 100 ms and 50 ms, and σ to 60 ms in both conditions (see Table 1).

Table 1. Data-generating ex-Gaussian parameters for the two simulated conditions.

Blur	μ	σ	τ	Expected RT
Clear	500	60	100	600
Low-Blur	550	60	50	600

We sample with replacement 10000 times to get the dataset we want to use.

```r
n_per_group <- 100

d <- data.frame(
  Blur = factor(
    rep(c("Clear", "Low-Blur"), each = n_per_group),
    levels = c("Clear", "Low-Blur")
  )
)

parameter_index <- as.integer(d$Blur)

d$rt <- brms.exgaussian::rexgaussian2(
  n = nrow(d),
  mu = parameters$mu[parameter_index],
  sigma = parameters$sigma[parameter_index],
  tau = parameters$tau[parameter_index]
)
```

Estimating Ex-Gaussian Parameters with `{easystats}`

The `{modelbased}` package, which is part of the `{easystats}` ecosystem, can be used to obtain condition-specific posterior estimates from a fitted `brms` model. The `estimate_means()` function returns estimated parameters for each condition, whereas `estimate_contrasts()` computes differences between conditions.

As shown in Table 1, the present data were generated so that the two conditions have the same expected reaction time but differ in the Gaussian and exponential components of the ex-Gaussian distribution.

Under the classical ex-Gaussian parameterization, the expected response is

$$E(RT) = \mu + \tau. \tag{1}$$

Thus, the 50-ms increase in μ for the Low-Blur condition is offset by a 50-ms decrease in τ. The expected RT is therefore 600 ms in both conditions.

Default `brms` Parameterization

We first fit the default `brms` ex-Gaussian model. In this parameterization, the parameter labeled `mu` represents the expected value of the full ex-Gaussian distribution rather than the location of its Gaussian component.

The exponential component is labeled `beta`. On the response scale, `beta` corresponds to the classical ex-Gaussian parameter τ:

$$\beta_{\texttt{brms}} = \tau_{\text{classical}}. \tag{2}$$

Thus, `beta` and τ are not substantively different quantities. Both represent the mean of the exponential component. The difference is primarily one of notation. The important difference between the default `brms` model and the classical parameterization concerns `mu`:

$$\mu_{\texttt{brms}} = \mu_{\text{classical}} + \tau. \tag{3}$$

In other words, the default `brms` parameterization uses `mu` for the expected value of the complete ex-Gaussian distribution, whereas the classical parameterization uses μ for the location of the Gaussian component.

Both the expected response and the exponential component are allowed to vary across conditions.

Prior Specification

The priors were selected to reflect the known scale of the simulated data while remaining sufficiently broad to allow the model to recover deviations from the generating values.

```r
priors_default <- c(
  prior(
    normal(600, 100),
    class = "Intercept"
  ),
  prior(
    normal(0, 50),
    class = "b"
  ),
  prior(
    normal(log(100), 0.5),
    class = "Intercept",
    dpar = "beta"
  ),
  prior(
    normal(0, 0.5),
    class = "b",
    dpar = "beta"
  )
)
```

The intercept prior for the primary `mu` submodel is

$$\mu_{\text{Intercept}} \sim \mathcal{N}(600, 100). \tag{4}$$

We center this prior at 600 ms because, in the default `brms` parameterization, `mu` represents the expected RT. The simulated expected RT for the reference condition is

$$500 + 100 = 600 \quad \text{ms} \ . \tag{5}$$

The standard deviation of 100 ms makes the prior informative about the plausible scale of RTs without tightly constraining the intercept to equal exactly 600 ms.

The prior on the `Blur` coefficient is

$$b_{\text{Blur}} \sim \mathcal{N}(0, 50). \tag{6}$$

This prior is centered at zero because, before observing the data, positive and negative condition effects are both plausible. A standard deviation of 50 ms allows substantial differences between conditions while placing less prior mass on implausibly large effects.

The `beta` parameter is positive and uses a log link. Its intercept is therefore modeled on the log scale:

$$\log(\beta_{\text{Clear}}) \sim \mathcal{N}(\log(100), 0.5). \tag{7}$$

The prior is centered at $\log(100)$ because the simulated value of the exponential mean for the reference condition is 100 ms. After applying the inverse-link transformation, the prior is centered approximately around

$$\beta_{\text{Clear}} = 100 \quad \text{ms} \ . \tag{8}$$

The standard deviation of 0.5 is on the log scale. It therefore represents multiplicative rather than additive uncertainty.

The prior on the effect of `Blur` on `beta` is

$$b_{\beta,\ \text{Blur}} \sim \mathcal{N}(0, 0.5). \tag{9}$$

Because this coefficient is also on the log scale, a value of zero represents no difference between conditions:

$$\exp(0) = 1. \tag{10}$$

The simulated change from 100 ms to 50 ms corresponds to the log ratio

$$\log\left(\frac{50}{100}\right) = \log(0.5) \approx -0.693. \tag{11}$$

The prior $\mathcal{N}(0, 0.5)$ places reasonable prior mass near this value while still favoring smaller differences.

The model can then be fit as follows:

```
fit_brms <- brm(
  bf(
    rt ~ Blur,
    sigma ~ 1,
    beta ~ Blur
  ),
  data = d,
  prior = priors_default,
  family = exgaussian(),
  chains = 2,
  iter = 2000,
  warmup = 500,
  backend = "cmdstanr",
  cores = 4,
  seed = 1234
)
```

Estimating the Expected Response

Condition-specific estimates of the expected response can be obtained with `estimate_means()` and setting `predict` argument to "mu":

```
overall_means <- estimate_means(
  fit_brms,
  by = "Blur",
   predict="mu",
  centrality = "mean"
)
```

Table 2. Estimated expected response, $E(RT)$, by condition from the default brms model.

Blur	Mu	95% CrI	pd
Clear	607.0	[584.2, 632.3]	1
Low-Blur	595.1	[580, 611.9]	1

Because the primary `mu` parameter in the default `brms` ex-Gaussian model represents $E(RT)$, these estimates should be close to 600 ms in both conditions.

The corresponding contrast can be obtained using `estimate_contrasts()`:

```
overall_difference <- estimate_contrasts(
  fit_brms,
  contrast = "Blur",
  centrality = "mean"
)
```

Table 3. Blur contrast in the expected response from the default brms model.

Level1	Level2	Mean	95% CrI	pd
Low-Blur	Clear	−11.9	[−40.6, 15.6]	0.796

The posterior estimate of this contrast should be close to zero because the two conditions were generated with the same expected RT:

$$(550 + 50) - (500 + 100) = 600 - 600 = 0. \tag{12}$$

Thus, the default model correctly recovers the absence of a difference in the full distributional mean. However, its primary `mu` coefficient does not directly reveal the 50-ms difference in the Gaussian-location parameter.

Estimating the Exponential Component

In the default `brms` parameterization, `beta` represents the mean of the exponential component. It therefore corresponds directly to τ in the classical parameterization:

$$\beta = \tau. \tag{13}$$

Condition-specific estimates can be obtained by requesting predictions for `beta`:

```
tau_estimates_default <- estimate_means(
  fit_brms,
  by = "Blur",
  predict = "beta",
  centrality = "mean"
)
```

Table 4. Estimated exponential component (`beta`) by condition from the default brms model.

Blur	Beta	95% CrI	pd
Clear	119.0	[91.9, 150.7]	1
Low-Blur	58.1	[38.6, 80.8]	1

Although the resulting object is called `tau_estimates_default` for interpretive clarity, the parameter is named `beta` inside the default `brms` model.

These estimates should be approximately 100 ms for Clear and 50 ms for Low-Blur.

The corresponding contrast can be obtained using:

```
tau_difference_default <- estimate_contrasts(
  fit_brms,
  contrast = "Blur",
  predict = "beta",
  centrality = "mean"
)
```

Table 5. Blur contrast in the exponential component (`beta`) from the default brms model.

Level1	Level2	Mean	95% CrI	pd
Low-Blur	Clear	−60.9	[−93.9, −28.8]	1

The expected contrast is

$$50 - 100 = -50 \quad \text{ms} . \tag{14}$$

Although the default model estimates the exponential component directly, it does not directly return the classical Gaussian-location parameter. That quantity must instead be derived as

$$\mu_{\text{Gaussian}} = E(RT) - \beta. \tag{15}$$

Because $\beta = \tau$, this can also be written as

$$\mu_{\text{Gaussian}} = E(RT) - \tau. \tag{16}$$

Classical Ex-Gaussian Parameterization

We can instead fit the model using the `{brms.exgaussian}` package, which implements the classical ex-Gaussian parameterization. In this model, `mu` represents the location of the Gaussian component and `tau` represents the mean of the exponential component.

Thus, the difference between the two models is not that `beta` and `tau` represent different components. Rather, the packages use different labels for the same exponential parameter:

$$\beta_{\texttt{brms}} = \tau_{\texttt{brms.exgaussian}}. \tag{17}$$

The main difference is the interpretation of `mu`:

$$\mu_{\texttt{brms}} = E(RT), \tag{18}$$

whereas

$$\mu_{\texttt{brms.exgaussian}} = \mu_{\text{Gaussian}}. \tag{19}$$

We first define the custom family and Stan functions:

```
exg_stanvars <- exgaussian2_stancode()
exg_family <- exgaussian2()
```

Prior Specification

The priors for the classical model are specified as follows:

```
priors_classical <- c(
  prior(
    normal(500, 100),
    class = "Intercept"
  ),
  prior(
    normal(0, 50),
    class = "b"
  ),
  prior(
    normal(log(100), 0.5),
    class = "Intercept",
    dpar = "tau"
  ),
  prior(
    normal(0, 0.5),
    class = "b",
    dpar = "tau"
  ),
  prior(
    normal(log(60), 0.5),
    class = "Intercept",
    dpar = "sigma"
  )
)
```

These priors follow the same reasoning as those for the default model (described above), with two changes. First, the intercept is now centered at 500 ms rather than 600 ms, because in the classical parameterization `mu` denotes the Gaussian-location parameter (500 ms for the Clear reference condition) rather than the expected RT:

$$\mu_{\text{Intercept}} \sim \mathcal{N}(500, 100). \tag{20}$$

Second, because this model estimates `sigma` explicitly, it receives its own weakly informative prior on the log scale, centered at the simulated Gaussian standard deviation of 60 ms:

$$\log(\sigma) \sim \mathcal{N}(\log(60), 0.5). \tag{21}$$

The coefficient for `Blur` on `mu` and the exponential-component priors are unchanged from the default model; the latter now simply attach to `tau` rather than `beta`, reflecting the difference in labels rather than any difference in the underlying quantity.

The model is then fit using:

```
fit_brms_new <- brm(
  bf(
    rt ~ Blur,
    tau ~ Blur,
    sigma ~ 1
  ),
  data = d,
  prior = priors_classical,
  family = exg_family,
  stanvars = exg_stanvars,
  chains = 2,
  iter = 2000,
  warmup = 500,
  backend = "cmdstanr",
  cores = 4,
  seed = 1234
)
```

Estimating the Gaussian-Location Parameter

For the classical model, estimates from the primary regression formula correspond to the Gaussian-location parameter μ:

```
mu_estimates <- estimate_means(
  fit_brms_new,
  by = "Blur",
  predict="mu",
  centrality = "mean"
)
```

Table 6. Estimated Gaussian-location parameter (μ) by condition from the classical model.

Blur	Mu	95% CrI	pd
Clear	490.4	[467.4, 514.5]	1
Low-Blur	535.4	[514.2, 556.6]	1

The estimates should be approximately 500 ms for Clear and 550 ms for Low-Blur.

The contrast can be obtained using:

```
mu_difference <- estimate_contrasts(
  fit_brms_new,
  contrast = "Blur",
  predict="mu",
  centrality = "mean"
)
```

Table 7. Blur contrast in the Gaussian-location parameter (μ) from the classical model.

Level1	Level2	Mean	95% CrI	pd
Low-Blur	Clear	45.1	[18.1, 71.7]	1

The expected contrast is

$$550 - 500 = 50 \quad \text{ms} \; . \tag{22}$$

The posterior mean will not necessarily equal exactly 50 ms because of sampling variability. However, the 95% credible interval should contain the population value of 50 ms when the model successfully recovers the simulated effect.

Estimating the Exponential Component

Condition-specific estimates of τ can be obtained by requesting predictions for the `tau` distributional parameter:

```
tau_estimates <- estimate_means(
  fit_brms_new,
  by = "Blur",
  predict = "tau",
  centrality = "mean"
)
```

Table 8. Estimated exponential component (τ) by condition from the classical model.

Blur	Mean	95% CrI	pd
Clear	118.2	[90, 151.3]	1
Low-Blur	58.8	[38.3, 83]	1

The estimates should be close to 100 ms for Clear and 50 ms for Low-Blur.

The corresponding contrast can be obtained using:

```
tau_difference <- estimate_contrasts(
  fit_brms_new,
  contrast = "Blur",
  predict = "tau",
  centrality = "mean"
)
```

Table 9. Blur contrast in the exponential component (τ) from the classical model.

Level1	Level2	Mean	95% CrI	pd
Low-Blur	Clear	−59.4	[−94.3, −26.1]	1

The expected contrast is

$$50 - 100 = -50 \quad \text{ms} \; . \tag{23}$$

Calculating the Expected Response

Under the classical parameterization, the expected response is not represented by either parameter alone. It must be calculated as

$$E(RT) = \mu + \tau. \tag{24}$$

For the Clear condition,

$$E(RT_{\text{Clear}}) = 500 + 100 = 600 \quad \text{ms} \; . \tag{25}$$

For the Low-Blur condition,

$$E(RT_{\text{Low-Blur}}) = 550 + 50 = 600 \quad \text{ms} . \tag{26}$$

The difference in expected RT is therefore

$$E(RT_{\text{Low-Blur}}) - E(RT_{\text{Clear}}) = 600 - 600 = 0 \quad \text{ms} . \tag{27}$$

Although the classical model should recover a positive blur effect on μ and a negative blur effect on τ, these effects cancel when the expected response is calculated. The full distributional mean therefore remains approximately equal across conditions.

This example illustrates the central difference between the two parameterizations. The default `brms` ex-Gaussian model directly estimates the expected response and labels that parameter `mu`. The exponential component is labeled `beta`, although it corresponds to the classical τ parameter. In contrast, the `{brms.exgaussian}` model uses `mu` for the Gaussian-location parameter and `tau` for the exponential mean, consistent with the classical parameterization commonly used in reaction-time research.

Conclusion

With this commentary, I hope to encourage researchers to use the appropriate ex-Gaussian parameterization when modeling reaction-time data with the {brms} package. The ex-Gaussian distribution provides a powerful framework for moving beyond analyses of the mean by separating differences in the central portion of the reaction-time distribution from differences in its right tail.

However, this flexibility also requires careful attention to model parameterization. In particular, researchers should not assume that parameters with familiar labels necessarily have the same interpretation across software implementations. Researchers should therefore examine the statistical definition and underlying implementation of a model before fitting it and interpreting its parameters. This is especially important when using flexible modeling software that may adopt parameterizations that differ from those commonly used in a particular research literature. I hope this commentary serves both as a practical guide to fitting the classical ex-Gaussian model in {brms} and as a broader reminder that convenient software defaults should not replace careful consideration of what a model actually estimates.

References

Balota, D. A., & Yap, M. J. (2011). Moving beyond the mean in studies of mental chronometry the power of response time distributional analyses. *Current Directions in Psychological Science*, *20*(3), 160166. http://cdp.sagepub.com/content/20/3/160.short

Bezanson, J., Edelman, A., Karpinski, S., & Shah, V. B. (2017). Julia: A fresh approach to numerical computing. *SIAM Review*, *59*(1), 65–98. https://doi.org/10.1137/141000671

Carpenter, B., Gelman, A., Hoffman, M. D., Lee, D., Goodrich, B., Betancourt, M., Brubaker, M. A., Guo, J., Li, P., & Riddell, A. (2017). Stan: A probabilistic programming language. *Journal of Statistical Software*, *76*(1), 1–32. https://doi.org/10.18637/jss.v076.i01

Geller, J., Gomez, P., Buchanan, E. M., & Makowski, D. (2025). A distributional response time analysis of the perceptual disfluency effect. *Journal of Cognition*, *8*(1), 50. https://doi.org/10.5334/joc.469

Matzke, D., & Wagenmakers, E.-J. (2009). Psychological interpretation of the ex-gaussian and shifted wald parameters: A diffusion model analysis. *Psychonomic Bulletin & Review*, *16*(5), 798817. http://link.springer.com/article/10.3758/PBR.16.5.798

Pfadt, J. M., Bartoš, F., Godmann, H. R., Waaijers, M., Groot, L., Heo, I., et al. (2025). A methodological metamorphosis: The rapid rise of bayesian inference and open science practices in psychology. *PsyArXiv*. https://doi.org/10.31234/osf.io/ck3js_v1

R Core Team. (2024). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. https://www.R-project.org/

Ratcliff, R., & Murdock, B. B. (1976). Retrieval processes in recognition memory. *Psychological Review*, *83*(3), 190–214. https://doi.org/10.1037/0033-295X.83.3.190

Van Rossum, G., & Drake, F. L. (2009). *Python 3 reference manual*. CreateSpace.

Vazire, S. (2018). Implications of the credibility revolution for productivity, creativity, and progress. *Perspectives on Psychological Science*, *13*(4), 411–417. https://doi.org/10.1177/1745691617751884

Wilcox, R. R., & Keselman, H. J. (2003). Modern robust data analysis methods: Measures of central tendency. *Psychological Methods*, *8*(3), 254274. https://doi.org/10.1037/1082-989X.8.3.254